Université d'Ottawa
Faculté des sciences

University of Ottawa
Faculty of Science

PHY 3902: Physics and Applied Physics Laboratory I

Experiment 4:

Servo Control Loops (Feedback Control Systems) – PID Temperature Controller

**** Work with a partner for this laboratory. You may submit one lab report per team.**

**** Your lab report should follow a standard format as described on Brightspace. Don't forget to address the various questions posed in this guide!**

Objectives:

- Understand PID feedback control systems
- Set and control the temperature of an object
- Analyze your results and achieve good STABILITY, ACCURACY and RESPONSE
- Quantify the specific heat of a material

Caution: You will be controlling a heating element to change the temperature of a metal block. DO NOT run your experiment unattended. Always keep track of the current temperature. Do not overheat the block. Do not touch the block when it is hot, as it is possible to burn yourselves. Include code to limit the temperature to below a maximum value of 75C.

***Caution:** electronic devices not connected to ground are very sensitive to static electricity and can be rendered completely and irreversibly inoperative following a static discharge. Avoid working in dry carpeted rooms and discharge yourself regularly by touching a massive metallic object, copper water pipe etc.*

Introduction

The ability to measure and control a parameter in a system is very important in many different applications. This is usually done as part of a servo control loop (feedback control system), where the parameter of interest is measured and compared to a certain desired set point, and the system is then "controlled" to bring the system closer to this desired set point. This is done in "feedback", and the process is repeated continuously to maintain the desired state. The thermostat in your home is part of a feedback system, where the house temperature is measured, and the thermostat decides whether to turn your furnace ON or OFF. A more complicated example would be Tesla's "self-driving" car, where many parameters (inputs) are measured and analyzed by software, while the car is controlled (outputs) to drive within the desired constraints (e.g. not driving into a tree!). It should be noted that many control systems are "analog" as opposed to "digital" (computer-controlled). However, while the discrete nature of

computer-controlled feedback systems can lead to reduced accuracy and control stability, using software does offer the advantage of keeping the control and data acquisition in one machine (e.g. computer, Arduino, etc.), and allows the feedback system to be tailored to different systems quite easily. Moreover, highly complex feedback behaviours can be programmed into the loop. Can you think of other feedback control systems?

In this laboratory, we will implement a temperature control system by measuring the temperature of a metal block (using a thermistor) and controlling a heater to adjust the temperature in order to achieve the desired temperature set point. This will be done within the main loop of the Arduino code, as the control process needs to be performed repeatedly if we want to maintain control of the system.

Concepts

The main objective of a temperature controller is to reach the desired temperature (set point) quickly, while minimizing overshoot, and then hold the set point as accurately as possible. Upon reaching a steady state, the heat energy provided by the heater will exactly balance the heat lost by the system to its surroundings. In reality, the amount of heat lost to the surroundings might vary, so a good feedback system should be able to react to changes in the environment. The main criteria that define good control systems are:

a. CONTROL ACCURACY (the temperature of the system should be close to the set point)
b. CONTROL STABILITY (fluctuation above and below the mean temperature should be small)
c. CONTROL RESPONSE (the system should adapt to changes in the set point rapidly and should reach steady state quickly)

Types of control

1. Open Loop "Control"

In some simple systems, a heater might be turned on continuously while the system temperature is allowed to come to equilibrium. This is not automated control as such, as the heater power remains unchanged except by human intervention. When cooking, the power delivered to the heating elements will be manually set and constant, and it is up to you to make sure your food does not burn!

2. On-Off Control

In this type of control system, the heater power is either fully ON or fully OFF. If the measured temperature is below the desired set point, the heater will be turned ON. If the measured temperature is above the desired set point, the heater will be turned OFF. That's it!

ON-OFF control system can be designed to have good accuracy and response. However, control stability is often not very good since the temperature must always cycle above and below the set point. The magnitude of these temperature fluctuations will depend on the system, its environment and the set

point. But in many systems, ON-OFF control is perfectly acceptable. Your electric oven probably uses ON-OFF control, and your chocolate chip cookies usually turn out fine….

However, ON-OFF control is inherently noisy and the heating element will be turned ON (100% power) or OFF (0% power) frequently, which is not acceptable in some applications. For instance, your home furnace (or air conditioning unit) might be damaged by frequent cycling. Or nearby sensitive electronic circuits might not behave properly if a heating element is cycled too often. A common compromise is to add HYSTERESIS to the control algorithm, so that once the set point is reached, the heater is turned off and does not turn on again until the measured temperature drops to a few degrees below the set point. A common approach to achieving this is to define a temperature BAND (range) around the set point. For example, for a set point of 55°C and a band of 2°C, the heater would be ON (100%) until the temperature reaches 56°C. The heater is then turned OFF (0%) to let the temperature drop to 54C before turning on again. This is represented in Figure 1.

This strategy will have the heater turn ON/OFF less frequently at the expense of some temperature variations. However, while the temperature in the above example will vary from 54°C to 56°C, the average temperature will be about 55°C, which is the set point. Home heating systems often include hysteresis to protect the heating or cooling systems.

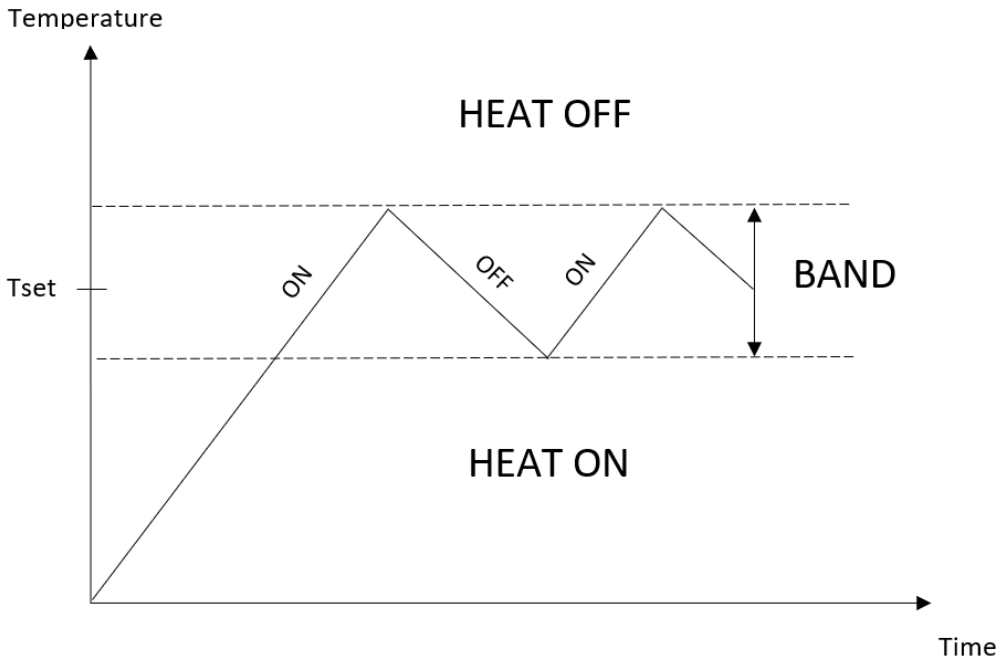


Figure 1: Hysteresis diagram of a residential thermostat

As can be seen in Figure 1, the output is either ON or OFF for any temperature in the range of the hysteresis, as determined per the direction of the curve.

3. PID control

PID control systems are more sophisticated as they can mitigate the issues related to simple ON-OFF control. PID is an acronym that stands for Proportional – Integral – Derivative. Each of these three components can be implemented independently, depending on the requirements.

3.1 Proportional Control

Proportional control can overcome the problem of temperature cycling and frequent switching of the heating element. This is achieved by continuously varying the heater power from 0% to 100% depending on the error (difference) between the set point and the measured temperature.

Implementation of proportional control is usually done within a TEMPERATURE BAND, within which proportional heater power is applied. As an example, let us assume a SET POINT of 55°C and a BAND of 10°C. When the measured temperature is below 50°C $\left(55°C - \frac{BAND}{2}\right)$, the heater power will be 100%. When the measured temperature is above 60°C $\left(55°C + \frac{BAND}{2}\right)$, the heater power will be OFF (0%). When the measured temperature is within the BAND (between 50°C and 60°C), the heater power would be set to be proportional to the difference (ERROR) between the desired SET POINT and the measured temperature. Mathematically, you can calculate the ERROR (E) as:

$$E = \frac{T_{set} - T_{measured}}{BAND}$$

and the heater power would be calculated to be a fraction of the maximum heater power (P_{max}):

$$Power = (0.5 + E) \times P_{max}$$

If you contemplate the above equations, you will realize that E can have a value between -0.5 and +0.5 (remember, we are only talking about measured temperatures within the band) and that the POWER will be somewhere between 0% and 100% of P_{max}.

But why did we add 0.5 in the power equation?! This was chosen as we assumed that at the SET POINT (E=0), the system will require 50% of the heater power to maintain the temperature at the SET POINT, in order to compensate for heat loss. This might seem like a reasonable assumption, but I assure you that it is likely quite wrong! If you supply 50% heater power when at the SET POINT, the system temperature is likely to either increase above or decrease below the SET POINT. This can be mitigated by reducing the BAND, but if the BAND gets too small, you will regain the temperature oscillations as the system gets closer to having ON-OFF control. By the way, you can add a proportional gain ($\boldsymbol{K_p}$) to the power equation for added control ($Power = \left(0.5 + K_p E\right) \times P_{max}$).

Proportional control leads to a stable temperature but accuracy is typically not great. One method of mitigating this lack of ACCURACY is to implement INTEGRAL CONTROL!

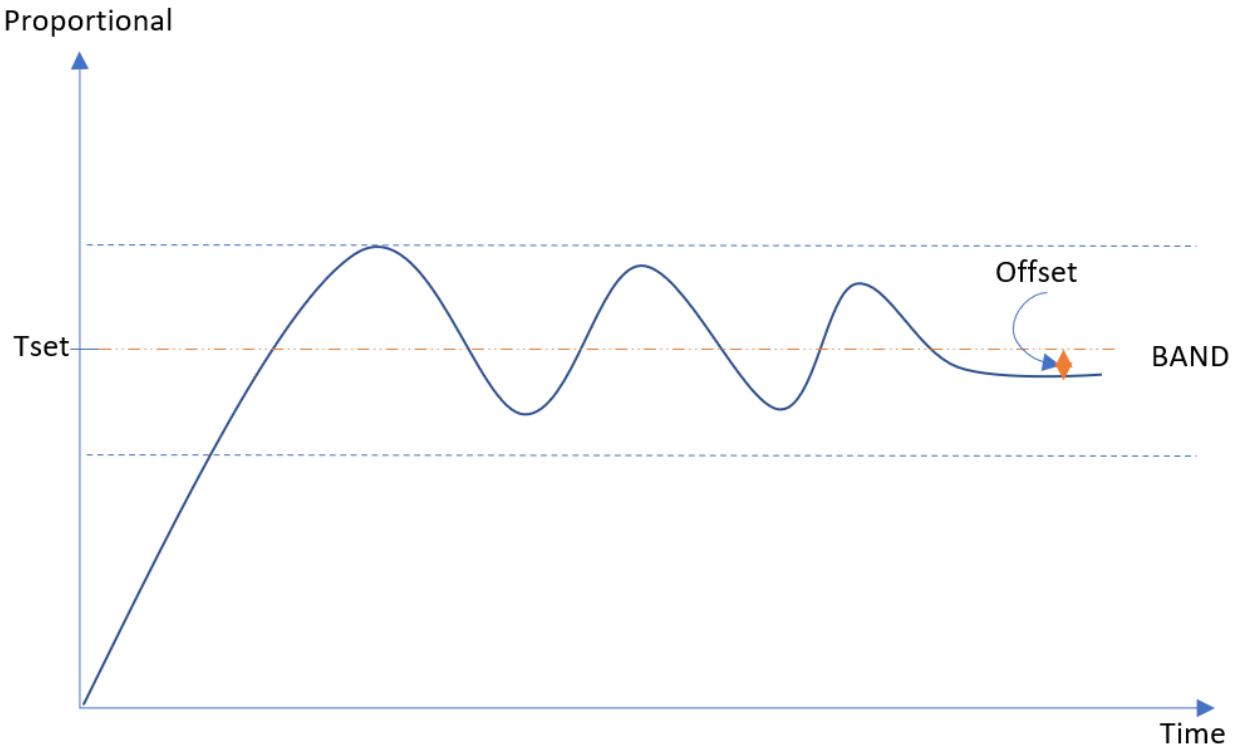


Figure 2: Temperature vs time for proportional control.

3.2 INTEGRAL CONTROL

Proportional control typically results in a stable temperature, but there is usually a residual error between the measured and desired temperature. An integrator term will be used to change the heater power to minimize this residual error.

To implement INTEGRAL CONTROL, you can calculate the integral of the error, which is the area between the measured temperature plot and the set point, as shown in Figure 3.

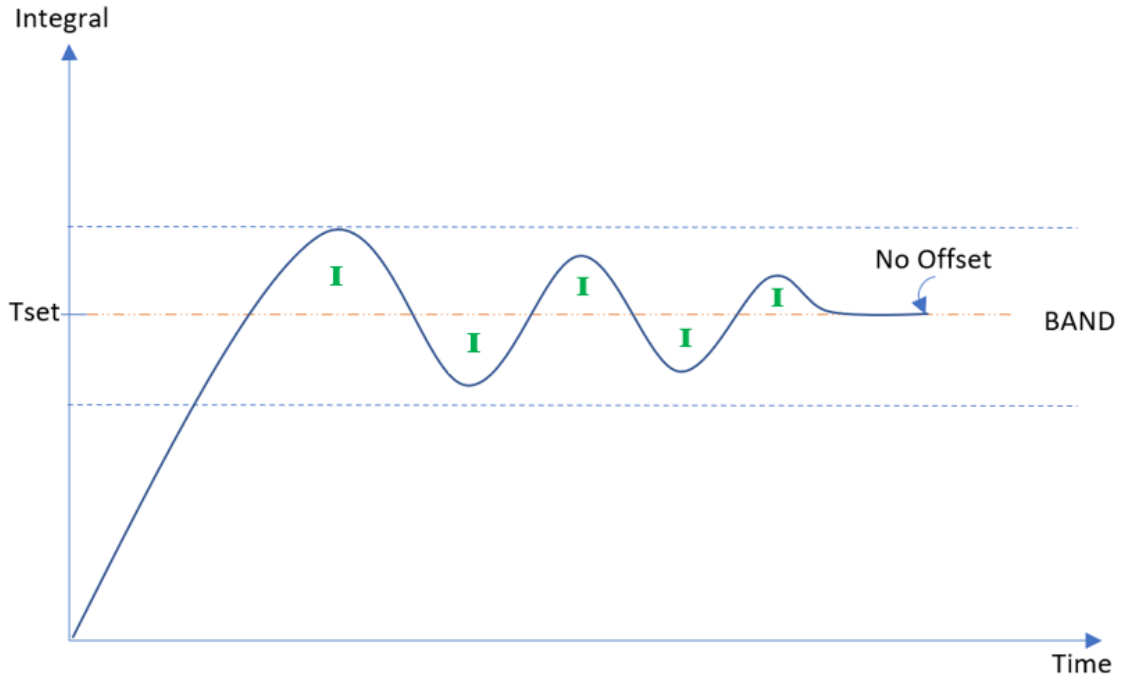


*Figure 3: Temperature vs time for integral control. The green "**I**" represents the areas between the measured temperature and the setpoint.*

The integral factor (I) can be calculated using:

$$I = \int E\, dt$$

where **dt** is the time interval between temperature measurements. The heater power then becomes:

$$Power = \left(0.5 + k_p E + k_I I\right) \times P_{max}$$

The constant $\boldsymbol{k_I}$ is added here to ensure that integral action is implemented slowly. Otherwise, additional temperature oscillations might be introduced. In my tests, I used $k_I \approx 0.00001$. You will certainly need to adjust this in your own systems.

If proportional control leads to the system stabilizing to a temperature below the desired SET POINT, then the integral term will gradually add power to the heater until the ERROR is zero, at which point the temperature will remain stable AND accurate.

One subtle caveat is that it is best to include an INTEGRAL ACTION TIME when programming integral control. This represents the time period over which the integral is calculated. This is usually about twice the response time of the system (the time it takes for the temperature to change by the value of the band). If you calculate the integral over larger time intervals, the system is likely to take a long time to stabilize. Think about why this is the case and comment on it in your report…

3.3 DERIVATIVE CONTROL

The combination of Proportional and Integral control is usually sufficient to achieving accurate and stable temperature control at a fixed set point. However, it is possible using DERIVATIVE CONTROL to improve the response of the system to sudden changes in the SET POINT. Without DERIVATIVE CONTROL, many systems will overshoot the desired SET POINT before stabilizing. The DERIVATIVE CONTROL monitors the rate of change of the measured temperature and modifies the control to reduce the rate of change when nearing the SET POINT.

The Derivative factor (D) can be calculated using:

$$D = -\frac{dE}{dt} = -\frac{\Delta E}{\Delta t}$$

And the power becomes:

$$Power = \left(0.5 + k_p E + k_I I + k_D D\right) \times P_{max}$$

where the constant $\boldsymbol{k_D}$ is used to control the influence of the derivative term. You can disable derivative control by making $\boldsymbol{k_D = 0.0}$ if you find that the DERIVATIVE CONTROL introduces undesired effects.

Like INTEGRAL CONTROL, DERIVATIVE CONTROL should also include an Action Time.

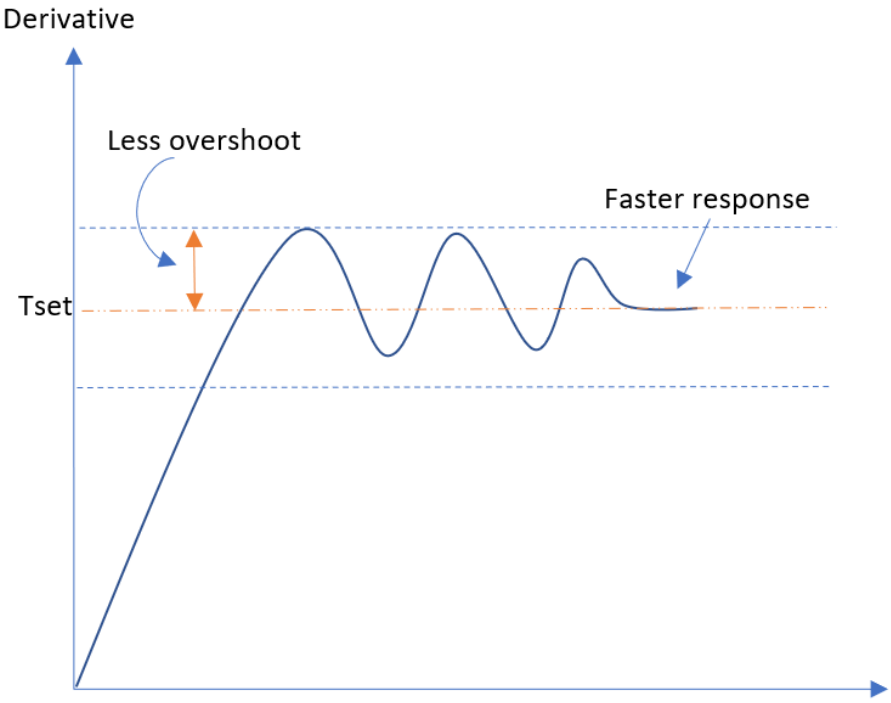


Figure 4: Temperature vs time for derivative control

Methods

Required components:

- ✓ Arduino Uno R3
- ✓ Thermistor NTC 3950 100 kΩ nominal at 25°C.
- ✓ Resistor 100 kΩ
- ✓ MOSFET transistor
- ✓ Heating element
- ✓ Metal block
- ✓ USB 2.0 cable to connect Arduino to host computer
- ✓ 12V wall adapter
- ✓ Wiring kit
- ✓ Prototyping board
- ✓ Arduino IDE installed.

You have access to a little aluminum block with some holes in it. The holes will allow you to insert a resistive heating element and a thermistor to measure the block's temperature. You will notice a third hole into which you can insert one of those little LEDs. You do not need this LED for this experiment, but if you wanted to assess the current-voltage characteristics of an LED as a function of temperature (see other experiment!), then you can! I hope that some of you try this!

THE THERMISTOR

In the INTRO experiment, you discovered how to measure the temperature using a thermistor as part of a voltage divider. Use the same strategy here. By the way, how would you calibrate the thermistor?

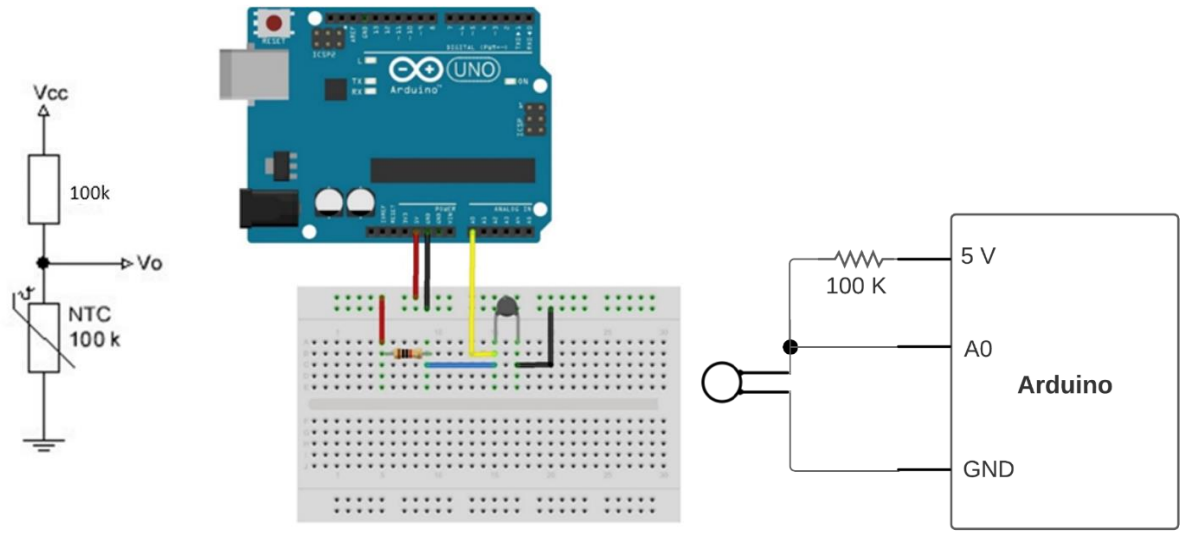


Figure 5: Voltage divider circuit with the thermistor

THE HEATER

The resistive heater becomes hot when you pass current through it. You will use the Arduino to output a voltage through the heating element. However, you should realize that the Arduino is not capable of providing enough power to truly heat the block. Therefore, you will use a separate power supply capable of outputting 12V at a maximum of 2A. You will use a MOSFET as a switch to allow this current to flow to the heating element. This will be controlled by the Arduino's PWM output voltage connected to the MOSFET's gate electrode. As an optional addition, you can consider connecting the same PWM output to a resistor (220 ohms) and an LED, in order to get a visual indication of the heater power.

I suggest you use one of the Arduino's PWM (Pulse Width Modulation) pins. In your Arduino code, you will use the function

analogWrite(MOSFET_PIN, PWM);

where "MOSFET_PIN" is the output PWM channel you decide to use and "PWM" will be a value between 0 and 255. This is the output range of the PWM output, where a value of zero corresponds to zero power and 255 corresponds to 100% power being sent through the heater. In **SETUP()**, do not forget to include the line:

pinMode(MOSFET_PIN, OUTPUT);

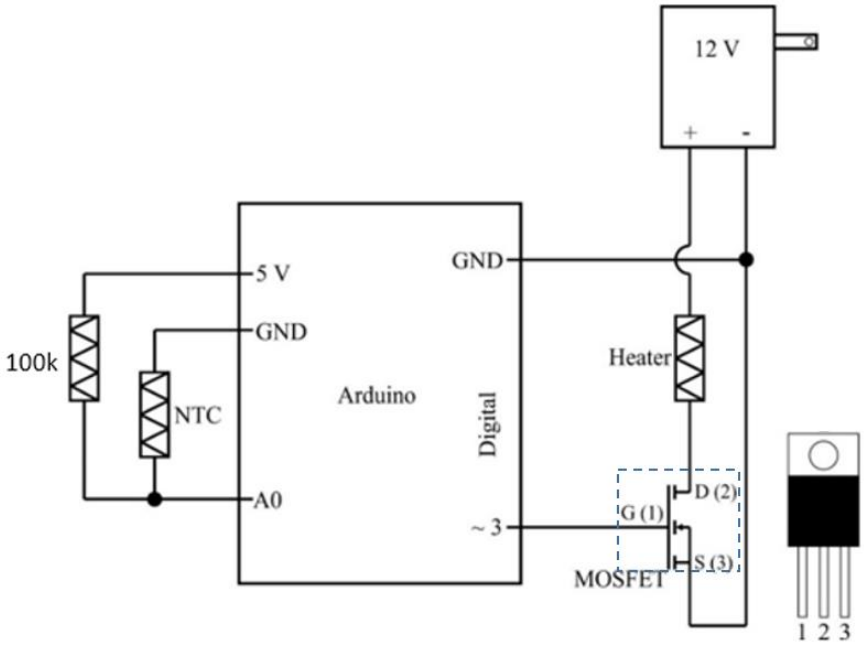


Figure 6: Temperature PID control schematic diagram

****** BE CAREFUL AS THE HEATER BLOCK CAN GET VERY HOT. ALWAYS UNPLUG THE POWER SOURCE TO THE HEATER WHEN NOT RUNNING AN EXPERIMENT. DO NOT LET YOUR BLOCK TO GO ABOVE 75°C.***

ON-OFF

As a starting point, you will have access to some basic Arduino code that implements basic ON-OFF temperature control. This file is named **ON_OFF.ino**. The feedback will be implemented within the main **LOOP()**. You should notice that the temperature is measured using the thermistor. Then, this measured temperature will be compared to the **SET POINT** and a decision is made either to turn ON the heater or turn it OFF. A time delay of 1000 ms is also imposed between iterations of the feedback loop. Finally, the measured temperature will be sent to the SERIAL MONITOR, where you can collect the data versus time for analysis.

You will notice that extra variables will be included in the code provided. These additional variables will be useful as you add features to the code. While you can certainly add variables or change variable names, using the recommended variables will make it a little easier for us when we look at your code!

You should change the SET POINT to see how the system behaves. I suggest keeping the SET POINT between 45°C and 65°C. Include these temperature versus time graphs in your reports and discuss your observations.

As a first modification, consider whether you should average multiple temperature measurements at every iteration in order to reduce noise, as was discussed in previous experiments.

Also, implement **HYSTERESIS** by using a **BAND**. Investigate how adding **HYSTERESIS** (and using different **BAND** values) affects stability and accuracy. Notice that the heating and cooling response times are different. This is likely to have an effect on control stability. Remember to show these temperature versus time graphs in your reports as well.

PROPORTIONAL CONTROL

Implement proportional control and investigate the effects of increasing the gain (reducing the BAND) on the stability, accuracy and response time. You are likely to notice a residual error between the SET POINT and the equilibrium temperature. Does your system respond similarly at different setpoints?

INTEGRAL CONTROL

Add INTEGRAL CONTROL to your temperature controller. Investigate how it eliminates the residual error observed when using proportional control alone. Does it allow for accurate control with larger BAND? You will certainly have to experiment with different time response constants to avoid oscillations, which is a common issue with PID control systems. What is the origin of this problem?

DERIVATIVE CONTROL

Does adding derivative control improve the system?!

QUANTIFYING THE SPECIFIC HEAT OF METALS

The specific heat of a substance is the amount of energy required to raise the temperature of 1 gram of the substance by 1°C. What is the difference between specific heat and heat capacity?

You can measure the resistance of the heating element. If you know the resistance and the voltage applied to the heater, you can determine the power supplied by the heater (P = VI). With this in mind, you can estimate the specific heat of the metal block by determining the power required to maintain different temperatures. You will also need to consider the cooling rate at those same temperatures. Compare your measurement to the actual heat capacity of Aluminum. Is your measurement comparable to expectations? What are your thoughts on any observed discrepancies? By the way, we should have a few extra blocks made of a different material. Can you figure out what metals they are made of from your measurements?!

COMMENTS

-Temperature control systems can be tested by choosing various SET POINTS and changing them suddenly. You can also analyse the response of the system when sudden cooling is applied to the block.

-Analysing your data:

- You should output the measured temperatures to the Serial Monitor, so that you can later copy/paste your data to a .CSV file and make corresponding graphs. A method for doing this has been discussed previously.
- However, it's also possible to use a Python script to extract data from the Serial Monitor in real-time in order to generate a real-time graph that will update automatically during the experiment. Information about doing this is on Brightspace. You can also choose to use Arduino's "Serial Plotter" to visualize your data in real-time.

**** Your lab reports should include several "temperature versus time" graphs that show the different behaviours of your systems under different types of control. You should discuss these differences and strategies for optimizing these types of control systems. Don't forget to copy/paste the various sketches you used for the different parts of this experiment (Hysteresis, Proportional, PID, etc.) at the end of your report.*

**** You should also plan to discuss the coding strategies you used to implement each type of control. While you will include your full code at the end of your reports, you can also discuss some of the details of this code in the report itself, as you see fit.*

IDEAS ON WHAT TO DISCUSS IN YOUR REPORTS:
You might notice that you can implement PID control pretty quickly, but you should ask yourself what

parameters are adjustable and how do they influence the response of the system. For instance, what happens when you change the band size? Do oscillations increase or decrease? Does the performance of your control system change when you use different set points? What about overshoot? It's common to notice that the measured temperature surpasses the set point early on. What can you do to prevent this? You can also adjust the constants (K_p, K_i, K_d) and the "action times". How does your system respond to sudden changes in temperature (some have used wet papertowel to apply sudden cooling to the metal bloc). Is it possible to achieve perfect control (reaching set point rapidly, no overshoot, no oscillations, etc.)?

As you will see, there are many parameters that can be adjusted to influence/optimize the performance of the system. You should be discussing your observations (with supporting data) in your lab reports. I know the lab manual does not provide specific details on what to measure and what to include in your lab reports. I know this is a little annoying. But we're trying to encourage you to think about the experiment you are conducting and about making decisions on how best to implement your own experiment, and how to present your observations. Real scientific research usually involves these types of decision-making steps.